

Minimum Requirements Document

Transceiver for SkyVision Test Site

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Background & Scope: The AFRL SkyVision Test Site (or outdoor laboratory) at Springfield-Beckley Airport in Springfield, OH required communication capability between air assets and monitoring of the local air traffic to maintain situations awareness and safe operations. AFRL Operations Directorate is responsible for approving, authorizing and monitoring S&T flight test activities and as such requires this radio capability to communication with both military and civilian aircraft. This requires a radio capable of both VHF and UHF communication. Additionally, the radio must be approved by the DAF, FCC and FAA.

Requirements/Salient Characteristics: Per DoD mandate, transceivers must meet spectrum clearance and authorization based on power output, antenna lobes frequency bleed, etc.

1. Software Defined transceiver capable of VHF and UHF AM Voice channels used by Commercial and Military Aircraft
 - a. UHF/VHF AM Voice: Channel Spacing: 25 kHz, 8.33 kHz, ETSI EN 300 676-1 Compliant
 - b. UHF/VHF FM Voice with optional 30-90 MHz frequency band extension
 - c. Frequency range: VHF: 115 MHz to 150 MHz (AM)
 - i. VHF: 115 MHz to 173.995 MHz (FM)
 - ii. UHF: 225 MHz to 399.995 MHz (AM and FM)
 - iii. LVHF: 30 MHz to 90 MHz (FM)
 - d. Tuning increments: 25 kHz, 12.5 kHz, 5 kHz and 8.33 kHz
 - e. Image response: ≥ 80 dB
 - f. Spurious response: ≥ 70 dB (≥ 80 dB typical)
 - g. Two-channel Remote Control Unit
 - h. Headset: Plantronics ENCOREPRO 540
 - i. Rack mounted capability
 - j. AC and DC capability (AC/DC Power Supply (10W systems))
 - k. Cabling
2. Associated antennas which also meet spectrum authorization requirements
3. Remote Control Unit and VoIP Gateway Adapter for long-distance operation over IP Networks: Must display channel and frequency currently tuned and ability to change channel/frequency at the control unit
4. MIL-STD-461G EMI/EMC Certified
5. Installation of 1 radio into Mobile Ground Based Detect and Avoid SkyVision Vehicle: Pictures Attached
6. Installation of 1 radio into Springfield-Beckley Airport Tower, Springfield, OH: Pictures Attached